

Multi-Subnet IPv6 Addressing & Inter-Interface Routing Lab Report

Prepared By: Eng: Mohamed Ashraf

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Technology: Cisco IOS (IPv6 Unicast Routing)

Environment: Cisco Packet Tracer

1. Executive Summary

As modern enterprise networks continuously transition away from IPv4 depletion limitations, mastering stateless and stateful IPv6 integration becomes standard practice. This technical lab environment demonstrates the absolute fundamentals of native IPv6 global unicast addressing (GUA), link-local structural behaviors, and standard multi-interface direct inter-subnet routing.

Using a centralized Cisco Layer 3 routing engine topology connected to separate isolated switching blocks, three distinct IPv6 subnets were deployed and configured under a standard /64 architectural boundary. Device connectivity verification was executed perfectly across the network matrix via standard ICMPv6 echo sequencing, verifying operational interface routing states.

2. Topology & Architectural Design

The infrastructure consists of one Cisco Router (**R1**) handling layer 3 traffic distribution across three discrete local area networks. Each sub-interface partition maps to a unique subnet parsed from the documentation allocation block.

Network Addressing Scheme

Network Name	IPv4 Network (Ref)	IPv6 Subnet Block	Router Interface	Interface IPv6 Address	End Device IP (PC)
Subnet 1 (LAN 1)	192.168.1.0/24	2001:DB8:0:1::/64	GigabitEthernet0/0	2001:DB8:0:1::1/64	2001:DB8:0:1::2/64
Subnet 2 (LAN 2)	192.168.2.0/24	2001:DB8:0:2::/64	GigabitEthernet0/1	2001:DB8:0:2::1/64	2001:DB8:0:2::2/64
Subnet 3 (LAN 3)	192.168.3.0/24	2001:DB8:0:3::/64	GigabitEthernet0/2	2001:DB8:0:3::1/64	2001:DB8:0:3::2/64

3. Device Configurations

To allow transit routing between separate interfaces on a Cisco Router for IPv6 traffic, the execution of the global unicast structural process is required before configuring explicit addresses.

Router R1 Interface Provisioning Commands

```
R1# configure terminal
R1(config)# ipv6 unicast-routing

R1(config)# interface gigabitEthernet 0/0
R1(config-if)# description LAN_1_Gateway
R1(config-if)# ipv6 address 2001:db8:0:1::1/64
R1(config-if)# no shutdown

R1(config)# interface gigabitEthernet 0/1
R1(config-if)# description LAN_2_Gateway
R1(config-if)# ipv6 address 2001:db8:0:2::1/64
R1(config-if)# no shutdown

R1(config)# interface gigabitEthernet 0/2
R1(config-if)# description LAN_3_Gateway
R1(config-if)# ipv6 address 2001:db8:0:3::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# exit
R1# copy running-config startup-config
```

4. Operational Verification & Analysis

Following configuration deployment, validation commands were executed on the control plane to ensure interfaces transitioned successfully to an active status and that the IPv6 routing database updated correctly.

4.1 Interface Brief Verification

Execution of `show ipv6 interface brief` displays active configurations. The system auto-generates EUI-64 or customized Link-Local addresses (prefixed with **FE80::**) uniquely corresponding to each interface alongside the explicitly defined Global Unicast Addresses:

```

R1# show ipv6 interface brief
GigabitEthernet0/0      [up/up]
    FE80::201:97FF:FE9A:AC01
    2001:DB8:0:1::1
GigabitEthernet0/1      [up/up]
    FE80::201:97FF:FE9A:AC02
    2001:DB8:0:2::1
GigabitEthernet0/2      [up/up]
    FE80::201:97FF:FE9A:AC03
    2001:DB8:0:3::1
Vlan1                   [administratively down/down]
    unassigned

```

4.2 IPv6 Routing Table Validation

The command `show ipv6 route` confirms that the Cisco IOS kernel has populated the local routing database with Connected (C) and Local (L) host routes for all attached transit segments:

```

R1# show ipv6 route
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external

C   2001:DB8:0:1::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L   2001:DB8:0:1::1/128 [0/0]
    via GigabitEthernet0/0, receive
C   2001:DB8:0:2::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L   2001:DB8:0:2::1/128 [0/0]
    via GigabitEthernet0/1, receive
C   2001:DB8:0:3::/64 [0/0]
    via GigabitEthernet0/2, directly connected
L   2001:DB8:0:3::1/128 [0/0]
    via GigabitEthernet0/2, receive
L   FF00::/8 [0/0]
    via Null0, receive

```

Technical Insight: Notice the local loopback entries designated with a /128 mask. These represent specific host addresses assigned directly to the router's physical interfaces, optimized for internal traffic processing, whereas the /64 entries define the broad subnet reachability scope.

5. End-to-End ICMPv6 Data Plane Testing

Data plane integrity was checked from the terminal command line of **PC1** (located inside the 2001:DB8:0:1::/64 block). Pings targeted remote destinations across the routed network layout.

ICMPv6 Traversal Diagnostics from PC1

```
C:\> ping 2001:DB8:0:3::2
Pinging 2001:DB8:0:3::2 with 32 bytes of data:
Reply from 2001:DB8:0:3::2: bytes=32 time=64ms TTL=127
Reply from 2001:DB8:0:3::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:0:3::2: bytes=32 time=15ms TTL=127
Reply from 2001:DB8:0:3::2: bytes=32 time=13ms TTL=127

Ping statistics for 2001:DB8:0:3::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 64ms, Average = 23ms

C:\> ping 2001:DB8:0:2::2
Pinging 2001:DB8:0:2::2 with 32 bytes of data:
Reply from 2001:DB8:0:2::2: bytes=32 time=11ms TTL=127
Reply from 2001:DB8:0:2::2: bytes=32 time=3ms TTL=127
Reply from 2001:DB8:0:2::2: bytes=32 time=13ms TTL=127
Reply from 2001:DB8:0:2::2: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:0:2::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 6ms
```

Conclusion

The 0% packet loss metric across multiple subnets confirms successful operational capability. The lab demonstrates basic structural knowledge of IPv6 subnet organization, interface link provisioning, and successful packet forwarding across a modern multi-segment environment.